
Routing Trajectories

Mark Oussoren
oussoren.mark@gmail.com

Formulating the Optimization Problem

Given a starting point $\vec{x}_0 = (x_0, y_0) \in \mathbb{R}^2$, an ordered sequence of circles to pass through with varying radii, $\{(\vec{v}_1, r_1), (\vec{v}_2, r_2), \dots, (\vec{v}_{n-1}, r_{n-1})\}$, and an end point $\vec{x}_n$, can you find the shortest (and longest) piecewise linear trajectory from $\vec{x}_0$ to $\vec{x}_n$?

This can be formally written as

$$\begin{aligned} \min \quad & \left\{ \sum_{i=1}^n \|\vec{x}_i - \vec{x}_{i-1}\|_2^2 \right\} \\ \text{s.t.} \quad & \|\vec{x}_i - \vec{v}_i\|_2^2 \leq r_i^2, \\ & x_i, y_i \in \mathbb{R} \end{aligned}$$

Now we'd like to reformulate this in matrix form to ideally plug into a solver. To do so, we want a matrix D such that $x_i \mapsto x_{i+1} - x_i$. The following matrix does exactly that

$$D = \begin{pmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Then, the objective function is given by $x^T P_0 x$ where $P_0 = D^T D$ (a PSD matrix). As for constraints, we know $(x_i - c_i)^2 = x_i^2 - 2c_i x_i + c_i^2$ which is of the form $x^T P_i x + q_i^T x + b_i \leq 0$ if we set P_i to be a matrix with 1 in the i th entry, q_i to be the row vector with $-2c_i x_i$ in the i th entry and 0 elsewhere, and b_i to be the column vector with $-c_i^2$ in the i th entry and 0 elsewhere. And so in summary, the optimization problem is given by

$$\begin{aligned} \min \quad & \{x^T P_0 x\} \\ \text{s.t.} \quad & x^T P_i x + q_i x + b_i \leq 0, \\ & x \in \mathbb{R}^{n+1} \end{aligned} \tag{1}$$

This is a QCQP where all the P_i are PSD, so a minimum can be found quickly with interior point methods. In 1, we do exactly that on some sample data using `cvxpy`.

If we try maximizing the above, most solvers would complain about not being able to maximize a convex function. This leads us to investigate how to approximate solutions for the longest trajectories.

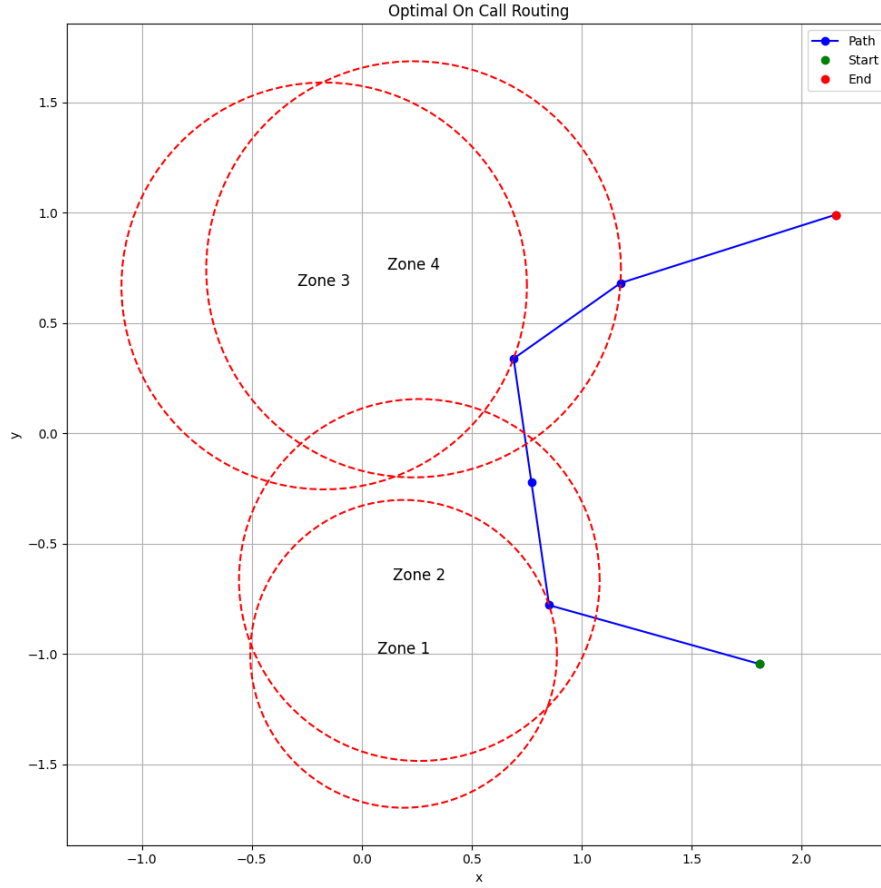


Figure 1: Sample Minimum Trajectory, $n = 5$

Reformulating the Non-Convex QCQP

If we instead want to maximize (1), then our objective function features a negative semidefinite matrix as $\max x^T A x = \min x^T (-A) x$. There are three primary means of solving this problem (to some ε -accuracy):

- Semidefinite Relaxation
- Branch and Bound
- Mixed Integer Linear Program (MILP) Reformulation

Constrained by time, we only apply semidefinite relaxation and sequential least squares programming (SLSP), implemented in scipy.

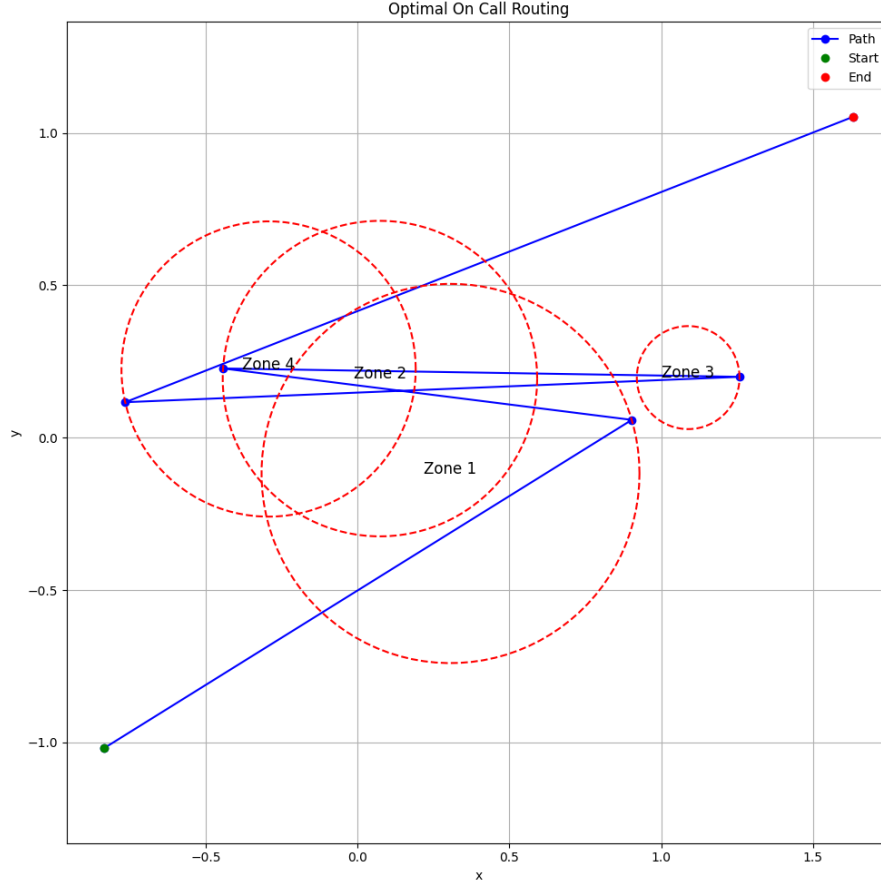


Figure 2: Sample Maximum Trajectory, $n = 5$

Finding Average Trajectories

Lastly, I'll briefly describe two means of approximating average trajectories.

1. Summing trajectories through the circle centers
2. Uniformly sampling from the uncertainty zones and averaging results (for some number of samples dependent on desired performance and accuracy)

Option (1) can be expressed as

$$\|\vec{v}_1 - \vec{x}_0\|_2^2 + \sum_{i=2}^{n-1} \|\vec{v}_i - \vec{v}_{i-1}\|_2^2 + \|\vec{x}_n - \vec{v}_{n-1}\|_2^2$$

while option (2) is similar, but we swap $\vec{v}_i = (x_i, y_i)$ with

$$\tilde{v}_i = (x_i + a_i \cos(\theta), y_i + a_i \sin(\theta))$$

where a_i is uniformly drawn from $[0, r_i]$ and θ_i is uniformly drawn from $[0, 2\pi)$ and we compute the average trajectory over $m \gg 0$ samples. Option (2) also provides us with estimates on the variance in route length.